

YANJING HUANG

yanjing.huang01@gmail.com | linkedin.com/in/yanjinghuang | yanjinghuang.github.io

EDUCATION

Carnegie Mellon University | *Master of Information Systems Management* **Pittsburgh, PA**
Coursework: Machine Learning for Problem Solving, Computational Data Science, *Aug 2024 – Dec 2025*
Distributed Systems, Object Oriented Programming in Java, Unstructured Data Analytics, and Database Management

Bucknell University | *Bachelor of Science in Business Administration* **Lewisburg, PA**
Major: Business Analytics | Minors: Computer Science, Economics *Aug 2018 – May 2022*

WORK EXPERIENCE

T-Mobile USA, Inc. **Bellevue, WA**
Assoc AI Modeling Engineer *Aug 2025 – Present*

- Designed and implemented an LLM-powered Feature Explorer Agent using RAG and LangGraph that generates model-ready feature tables; reduced feature discovery/selection time from 2 weeks to under 2 minutes based on user-defined modeling intent descriptions
- Optimized Retrieval-Augmented Generation (RAG) pipeline using FAISS vector database with 50K+ embeddings, implementing intelligent caching and pre-loading strategies that reduced index load time
- Architected and deployed FastAPI-based microservices on Azure App Service, integrating async I/O, agent pooling, and thread executors to reduce response latency by 70% under concurrent loads

Data Science Intern *May 2025 – Aug 2025*

- Built a DeepFM-based recommendation system with PyTorch and PySpark, modeling low- and high-order feature interactions across millions of user events. Achieved 88% Hit@5 by optimizing BPR loss, hybrid negative sampling, and attention-based features, while addressing cold-start users
- Built an LLM-driven feature mapping tool that automatically generates feature metadata files for the Feature Store, reducing manual mapping effort by 98% and yielding a cost saving of \$84K dollars
- Analyzed large-scale customer care transcripts using NLP techniques (text classification and clustering) to identify misrouted support cases, improving training data quality and increasing classification accuracy by 12%

Deloitte & Touche LLP **New York, NY**
Cyber & Strategic Risk Analyst (AI/ML Projects) *July 2022 – Aug 2024*

- Collaborated with engineering team to build and test a generative AI chatbot for a third-party risk management program, driving feature ideation, defining parameters, and executing testing to enhance functionality
- Built and launched an operational metrics dashboard highlighting managed risk, cost efficiency, and team performance gaps, providing client leadership with actionable insights for strategic decision-making
- Led client's third-party risk management program support page, spearheaded targeted training initiatives, and onboarded diverse teams

RESEARCH EXPERIENCE & PROJECTS

Auditing Zillow Housing Price Predictive Models | *Research Project* *Jan 2022 – May 2022*

- Built a custom **web crawler** using Python and **BeautifulSoup** to extract 10K+ property listings from Zillow, improving understanding of features influencing housing price estimation
- Engineered 20+ features and implemented **Random Forest** and **Linear Regression** models to estimate housing prices across diverse neighborhoods, identifying inaccuracies and biases in Zillow's predictive model

Winning the Digital Shelf | *Champion Analytics Case Competition* *Sept 2021 – Oct 2021*

- Placed 2nd among 12 university teams** in a retail analytics competition focused on optimizing digital shelf performance and customer behavior prediction
- Analyzed **50K+ transactions** in Python, using PCA and correlation analysis to identify **15 key features**, reducing model complexity by **60%**
- Built and tuned ensemble models (Random Forest, KNN, XGBoost) in scikit-learn, achieving **78% prediction accuracy** and delivering data-driven product placement recommendations

SKILLS

Tools & Frameworks: Databricks, Apache Spark, Azure ML, Git, Docker, MLflow, FastAPI, LangChain, Tableau
Programming: Python (NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, PyTorch, PySpark), Java, SQL, C, R
Skills: Machine Learning, Feature Engineering, Deep Learning (CNN, RNN, Transformers), Large Language Models (LLMs), Prompt Engineering, Retrieval-Augmented Generation (RAG), MLOps, REST APIs, Vector Databases (FAISS)